Optimized Design Parameter Dataset for Brushless DC Motor with Machine Learning-Based Validation

Methodology & Validation

1. Experimental Design, Materials and Methods

The data generation method involved writing a code in Python to compute the motor design parameters using the fundamental electromechanical equations. Python (NumPy / Pandas / Matplotlib) was used to perform numerical computations, statistical & variational analysis and visualization. Regression models were used to cross validate the generated motor design parameters. The mathematical and theoretical foundation for the study was based on standard BLDC motor design equations, supported by IEEE and IEC standard references to ensure accuracy and compliance with established design practices.

1.1. Computational Framework

The data generation process was automated using a Python script that implements the radial flux topology design equations. The process flow is described below:

Step 1: Computation of Motor Parameters

The design loop begins by defining ranges for independent variables:

- Rated Power (P_{hp}): 1 kW to 100 kW
- Rated Speed (S_r): 1000 to 6000 RPM
- Poles (N_m): 2, 4, 6, 8
- Magnetic Remanence (B_r): 1.2 T (Neodymium magnets)

Step 2: Geometric Sizing The mechanical speed (ω_m), electrical speed (ω_e), specific magnetic loading (B_g) and electric loading were utilized to determine the main dimensions (D^2L sizing equation).

Step 3: Electromagnetic Calculation The phase current (I_{ph}) required to produce the target torque (T) is derived using the torque equation and then maximum back EMF is computed using the distribution, pitch, and skew factors.

Step 4: Loss Calculation

- **Ohmic Loss (P_r):** Calculated based on the phase resistance (R_{ph}), which accounts for slot dimensions and end-turn length.
- **Core Loss (P_{cl}):** Modeled as a function of flux density and frequency using manufacturer loss data for electrical steel.

1.2 Descriptive Statistical Analysis

The descriptive statistics methods such as mean, median, minimum, maximum, standard deviation, skewness and kurtosis were used to investigate the reliability, distribution nature and physical possibility of the derived design variables of BLDC motor. Statistical measurements describe mean, standard deviation, variance, skewness, and excess kurtosis of important performance metrics, e.g. ohmic loss, core loss, phase current and efficiency over normalized power bins (Tables 1–8).

- **Loss Distribution:** The statistics of ohmic loss (Tables 1 and 2) presents a monotonous increase in means with power bins, from about 14 W in the smallest bin to 71 W in the largest bin. The small standard deviation within bin denotes that the dispersion is controlled and that it is a systematic scaling of the current and the conductor sizes. Bias values are close to zero ranging from −0.31 to 0.72, which reveals that none of the distributions has a significant asymmetry/ skewness pointing towards the tails of the distribution. The excess kurtosis is uniformly negative (≈ −1.2), which is another characteristic of platykurtic distribution, which simply means that the analytical variability is smooth rather than noisy for data is higher measurement. Conversely, mean magnitudes are at least an order of magnitude larger for core loss (Tables 3 and 4), ranging from 109 W at low power bins to 729 W at high power bins. It confirms the predominance of iron losses in high-speed permanent magnet machines which differentiates BLDC motors from induction motors where copper losses usually account for a bigger portion of total losses. The steady increase in variance unit across bins is expected since electrical frequency and magnetic loading scale approximately linearly. The negative kurtosis values support the outputs of the analytical solution are stable due to no abnormal accumulation of solutions therein.

- **Phase Current:** The statistical distributions of phase current grouped by torque bins (Tables 5 and 6) exhibit an expected growth in mean current for higher torque demands. The lower torque bins have almost normal distribution properties (skewness ~= 0) whereas the higher torque bins are bounded by the maximum current density imposed in the optimisation, and is therefore somewhat positively skewed. The trend of the variances shows that the electromagnetic loading scales perfectly physically with output torque, which proves the torque–current proportionality embedded in the BLDC motor design equations.

- **Efficiency:** The efficiency statistics (Table 7 and 8) results show a distribution tightly clustered around 95%, with very low standard deviation (< 0.18) in all bins. The slight difference is a testament to the strength of the optimization method to generate high-efficiency designs at various operating points. Since the value of Skewness was close to the zero, the data were approximately symmetrically distributed and they had no extreme outliers. Unlike experimentally measured data sets, where efficiency may vary with temperature or due to manufacturing tolerances, the present data set is analytical in nature and thus provides a controlled dispersion and deterministic consistency.

For all the parameters under study, the skewness is near 0 and the excess kurtosis is negative (approximately −1.0 to −1.2), indicating that the distributions are symmetric and platykurtotic. These characteristics show that the generated data are controlled by systematic electromechanical relations rather than noise. The monotonic scale observed in losses, current and torque by power bin is also a good physical indicator of the dataset. Similar to the previously published dataset for induction motors quantum of the above statistical measures are not just descriptive, but investigative in nature, confirming that the BLDC dataset is self-consistent and behaves as expected from an electromagnetic design perspective. As a result, the data set can be used for surrogate modelling, performance prediction, and machine learning-assisted validation.

Table 1

Descriptive statistics of Ohmic Loss (P_r) across normalized power bins, showing low variance and mean values for 6000 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	14.269935	30.969879	44.475514	57.285586	71.157919
sum	17123.922439	37163.854802	53370.616305	68742.703519	85389.503253
min	1.684135	23.453185	37.910806	50.954690	63.564222
max	23.437633	37.906536	50.944294	63.558165	92.282256
count	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
median	14.959753	30.809940	44.577234	57.412174	70.507806
std	6.019035	4.192263	3.809354	3.631667	5.284109
var	36.228788	17.575067	14.511181	13.189002	27.921810
Skewness	-0.314625	-0.050942	-0.003235	-0.044627	0.723673
Excess Kurtosis	-1.009601	-1.199208	-1.180734	-1.204347	0.213123
Range	21.753498	14.453351	13.033488	12.603475	28.718034

Table 2

Descriptive statistics of Ohmic Loss ($\boldsymbol{P_r}$) across normalized power bins, showing low variance and mean values for 200 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	15.581708	34.076452	49.166466	63.129213	76.515716
sum	623.268314	1363.058064	1966.658625	2525.168509	3060.628648
min	2.183145	26.061261	42.074056	56.446970	70.060641
max	25.621244	41.699701	56.098878	69.726425	82.919525
count	40.000000	40.000000	40.000000	40.000000	40.000000
median	16.251825	34.179793	49.209180	63.151996	76.529475
std	6.685022	4.676631	4.201000	3.979334	3.853880
var	44.689524	21.870875	17.648397	15.835101	14.852394
Skewness	-0.293067	-0.054422	-0.024837	-0.013954	-0.008692
Excess Kurtosis	-1.007385	-1.193154	-1.199240	-1.200583	-1.201044
Range	23.438100	15.638440	14.024823	13.279455	12.858884

Table 3

Descriptive Statistics of Core Loss ($\boldsymbol{P_{cl}}$), illustrating the dominance of iron losses ($> \mathbf{100}$ W) compared to Ohmic losses in high-speed operations for 6000 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	109.011353	308.970578	471.850481	603.919287	729.431677
sum	130813.624026	370764.693633	566220.576876	724703.144765	875318.012683
min	1.307538	217.596696	398.082835	544.213274	661.978558
max	217.553274	397.935160	544.064222	661.908565	889.678451
count	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
median	108.887263	311.528549	472.992920	603.927299	722.253036
std	64.024468	51.819809	42.500585	34.456402	45.730870
var	4099.132467	2685.292648	1806.299723	1187.243642	2091.312446
Skewness	-0.014914	-0.074104	-0.028087	-0.014862	0.644496
Excess Kurtosis	-1.223543	-1.132764	-1.237070	-1.232686	-0.172624
Range	216.245736	180.338464	145.981387	117.695290	227.699893

Table 4

Descriptive Statistics of Core Loss ($\boldsymbol{P_{cl}}$), illustrating the dominance of iron losses ($> \mathbf{100}$ W) compared to Ohmic losses in high-speed operations for 200 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	77.186009	227.826339	350.228477	447.790456	525.571968
sum	3087.440364	9113.053562	14009.139067	17911.618236	21022.878719
min	0.421622	158.905641	295.150011	403.907718	490.573992
max	155.108630	292.101497	401.479377	488.637753	558.150979
count	40.000000	40.000000	40.000000	40.000000	40.000000
median	77.500057	229.080774	351.258624	448.607134	526.222780
std	47.320217	39.926341	31.862302	25.389587	20.250307
var	2239.202899	1594.112702	1015.206286	644.631144	410.074953
Skewness	-0.005016	-0.076259	-0.078631	-0.078243	-0.078173
Excess Kurtosis	-1.236305	-1.195868	-1.194747	-1.194776	-1.194865
Range	154.687008	133.195856	106.329366	84.730035	67.576987

Table 5

Phase Current ($\boldsymbol{I_{ph}}$) Statistics grouped by Torque, exhibiting positive skewness consistent with high-torque demands for 6000 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	40.999545	100.198438	142.371271	170.716948	202.628152
sum	49199.454325	120238.126062	170845.525220	204860.337942	243153.782406
min	4.504761	60.847168	99.387258	131.806305	158.773355
max	87.298810	142.925933	185.628243	312.479033	366.647546
count	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
median	41.831668	102.606981	141.906472	168.697389	195.483736
std	18.776615	20.303370	14.684721	19.400491	34.864684
var	352.561260	412.226846	215.641044	376.379055	1215.546208
Skewness	-0.169041	-0.130162	0.085320	2.578105	2.943948
Excess Kurtosis	-0.949431	-1.125872	-0.027996	15.082061	9.369206
Range	82.794049	82.078765	86.240985	180.672728	207.874191

Table 6

Phase Current ($\boldsymbol{I_{ph}}$) Statistics grouped by Torque, exhibiting positive skewness consistent with high-torque demands for 200 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	40.748464	89.362523	147.370200	199.209190	228.037153
sum	1629.938543	3574.500929	5894.807988	7968.367584	9121.486101
min	4.527230	76.501542	101.785410	184.670826	214.200820
max	75.763818	101.218050	183.908122	213.481585	241.718747
count	40.000000	40.000000	40.000000	40.000000	40.000000
median	44.228161	89.625779	168.704081	199.280385	228.078660
std	19.752709	7.376744	34.547216	8.631977	8.246646
var	390.169501	54.416346	1193.510130	74.511030	68.007162
Skewness	0.101856	-0.088431	-0.369045	-0.020116	-0.012259
Excess Kurtosis	-0.727043	-1.183811	-1.793759	-1.199981	-1.200794
Range	71.236588	24.716509	82.122712	28.810759	27.517928

Table 7

Efficiency ($\boldsymbol{\eta}$) Statistics grouped by Total Losses, confirming a stable optimization target of $\approx \mathbf{95}\%$ for 6000 samples.

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	95.073988	95.114976	95.210950	95.288484	95.328503
sum	114088.785529	114137.971151	114253.139649	114346.180250	114394.203983
min	94.468936	94.537591	94.636538	94.736385	94.709181
max	95.637279	95.725224	95.734829	95.783644	95.841567
count	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
median	95.075129	95.116926	95.216498	95.288139	95.326602
std	0.183455	0.173769	0.177491	0.175638	0.164172
var	0.033656	0.030196	0.031503	0.030849	0.026952
Skewness	0.015162	-0.007896	-0.043351	-0.005552	-0.040722
Excess Kurtosis	-0.139025	0.065602	-0.138399	-0.057853	0.178693
Range	1.168343	1.187634	1.098291	1.047259	1.132386

Table 8

Efficiency ($\boldsymbol{\eta}$) Statistics grouped by Total Losses, confirming a stable optimization target of $\approx \mathbf{95}\%$ for 200 samples

bin	0-20%	20-40%	40-60%	60-80%	80-100%
mean	95.377945	95.373720	95.421521	95.457249	95.506807
sum	3815.117811	3814.948797	3816.860839	3818.289968	3820.272262
min	95.115516	95.077943	95.102788	95.110074	95.289309
max	95.636043	95.631883	95.983575	95.820816	95.880521
count	40.000000	40.000000	40.000000	40.000000	40.000000
median	95.391450	95.365303	95.425556	95.475078	95.509077
std	0.140116	0.138820	0.174114	0.150272	0.132491
var	0.019632	0.019271	0.030316	0.022582	0.017554
Skewness	-0.083403	-0.104899	0.658957	-0.080393	0.367374
Excess Kurtosis	-0.732638	-0.691386	1.222030	0.322883	-0.007446
Range	0.520527	0.553940	0.880787	0.710742	0.591211

1.3 Variational Analysis

In addition to descriptive statistical validation, a variational analysis was carried out to explore the physical relationships embedded within the dataset. This investigation demonstrates the predictable scaling relations among geometric parameters, electromagnetic quantities and performance parameters in the radial flux BLDC design space.

Figure 1 represents the influence of the number of turns/slot and the effective conductor cross-section area on the phase resistance. The graph shows a nonlinear increasing resistance as a function of turns per slot which is physically consistent considering that the conductor resistance is proportional to the length. As the number of turns is increased, the total length of copper increases linearly, and so do the resistive losses. On the other hand, increasing conductor area decreases resistance since resistance is inversely proportional to the cross-sectional area. The surface plot confirms the compromise between the generation of electromagnetic voltage (higher turns) and the least copper losses (large conductor area). The graded gradients in the plot also serve to confirm the purely deterministic based implementation of winding resistance equations used within the data set.

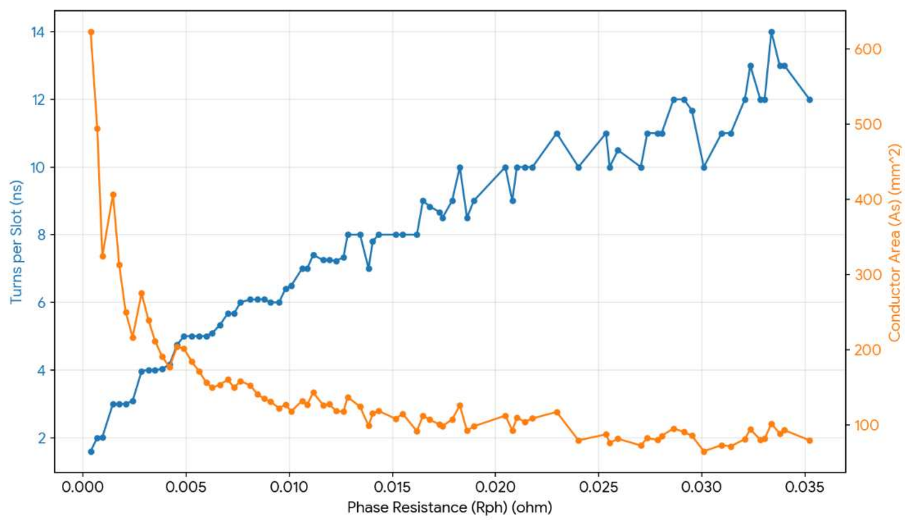

Fig. 1. Variation of phase resistance with respect to turns per slot and conductor area.

Figure 2 shows the dependence of the airgap flux on the pole pitch and on the stack length. The graph exhibits a universal-like behavior in which the airgap flux grows with stack length due to the increasing effective magnetic cross-sectional area. Also, the variations of pole pitch modify the airgap magnetic loading distribution. Increasing the pole pitch increases the flux linkage per pole, since more flux is gathered, while the flux density keeps within allowable magnetic limits. There is no sudden saturation of flux region in the plot which verifies that those design variables beyond the magnetic saturation threshold were filtered out during dataset validation. The smooth and physically sensible behaviour can be interpreted as an indication that magnetic loading constraints are properly applied in the analytical model.

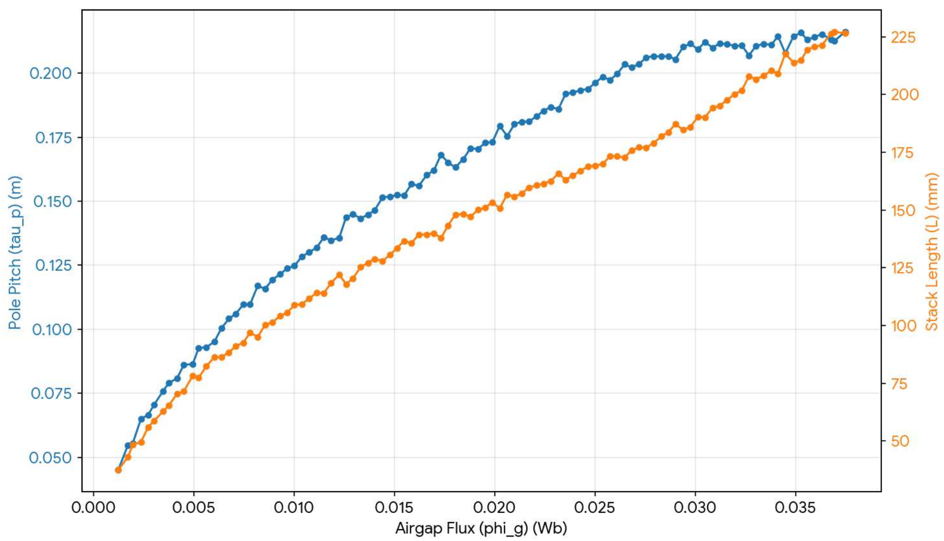

Fig. 2. Variation of airgap flux with respect to pole pitch and stack length.

Figure 3 illustrates the quadratic dependency of ohmic (copper) loss on phase current, consistent with the classical relation $P_{cu} = I^2R$. The plotted trend illustrates that copper loss rises nonlinearly as phase current increases, with resistance acting as a proportional scaling factor. The curvature shows that current magnitude has a dominant influence on loss increase compared to negligible variations in resistance. This behavior becomes predominantly substantial in high-torque operating regions where current density reaches its upper design limits. The smooth and continuous tendency observed in the figure authorizes the internal consistency of the analytically generated current–resistance–loss relationship within the dataset.

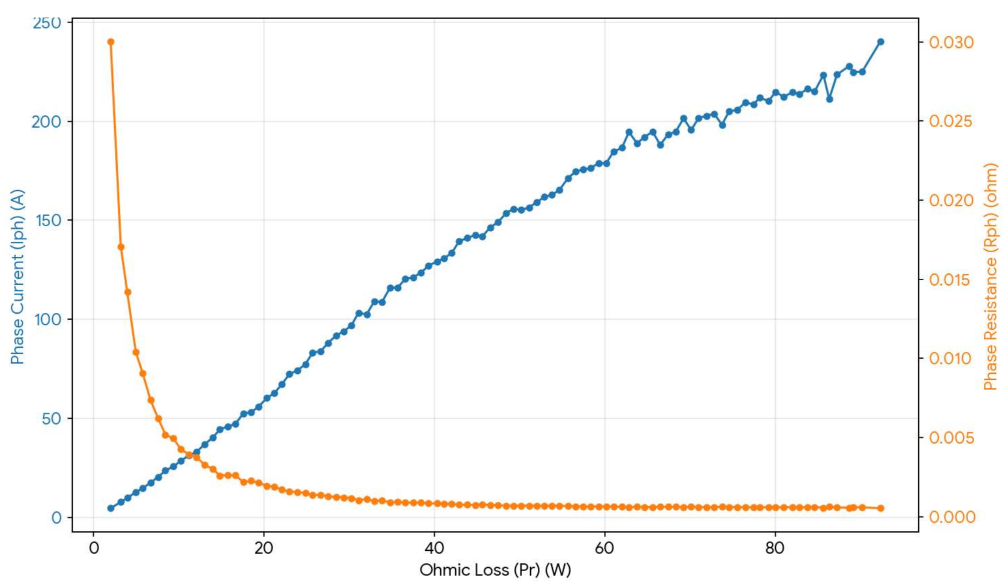

Fig 3. Variation of ohmic loss with respect to phase current and phase resistance.

Figure 4 shows the effect of full load losses on the efficiency at various power levels. The figure shows that efficiency is closely clustered within a tight range (around 95%) despite the growing output power. As expected, efficiency reduces gradually as the total losses rise; however, the slope is flat due to the well-optimized trade-off between copper and iron losses. The small curvature illustrates that efficiency reduction is not sudden and instead follows an expected scaling form. This stability authorizes the optimization problem encoded in the data generation process and authorizes that high efficiency designs are sustained throughout the operating regime.

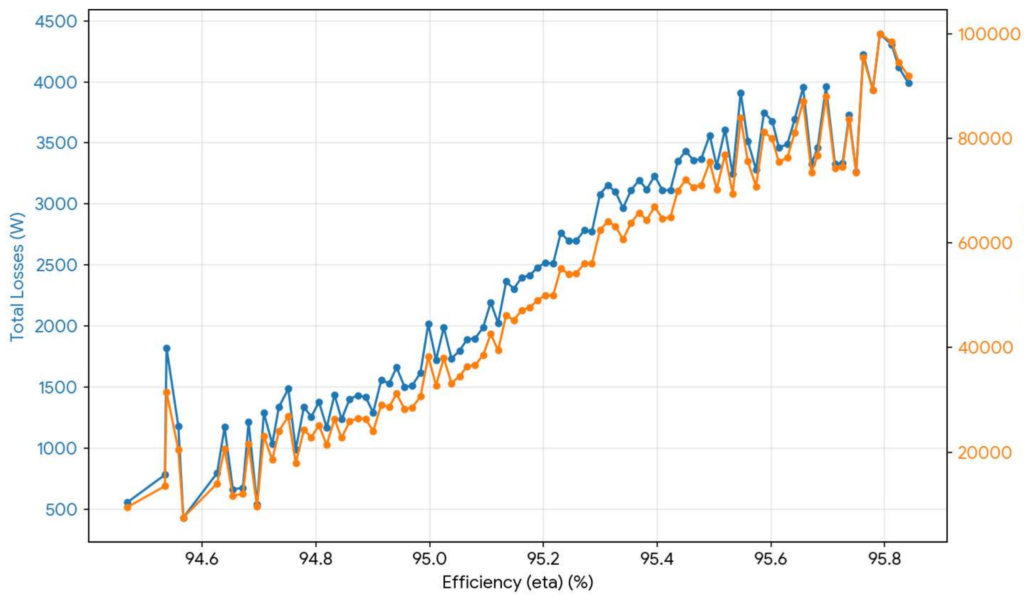

Fig 4. Variation of efficiency with respect to total full-load losses and output power.

Figure 5 represents the rated torque as a function of rated phase current, based on the electromagnetic torque equation of BLDC machines. The plot verifies the scaling of the phase current when the torque demand increases is nearly linear, and it is limited by the current density. Meanwhile, the back EMF presents a synchronized torque-dependent variation relating the magnet's flux linkage and speed. This correlation shows that the voltage balance equations are correctly applied in the analytical model of the equations. The fact that no sudden jumps were observed also supports deterministic torque–current–EMF coupling in the entire data set.

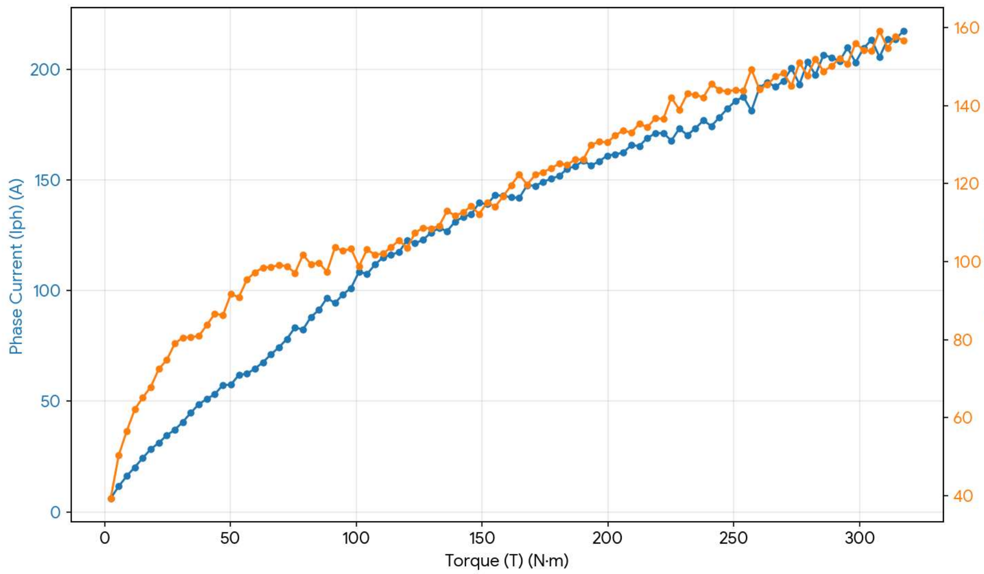

Fig 5. Variation of phase current and back EMF with respect to rated torque.

The reduction in contribution with load in core and ohmic losses is shown in Figure 6. The plot evidently shows that the core losses are the dominant part for the high-power bins, which is a typical feature of permanent magnet synchronous topologies implemented at high electrical frequency. Ohmic dissipation tops out lower and grows more slowly. The near linear relationship between total loss and core loss suggests that iron loss is the largest contributor to the efficiency bottleneck for this radial flux BLDC dataset. This trend sharply contrasts with the induction motor data sets, in which copper losses do account for a significant portion of the total losses.

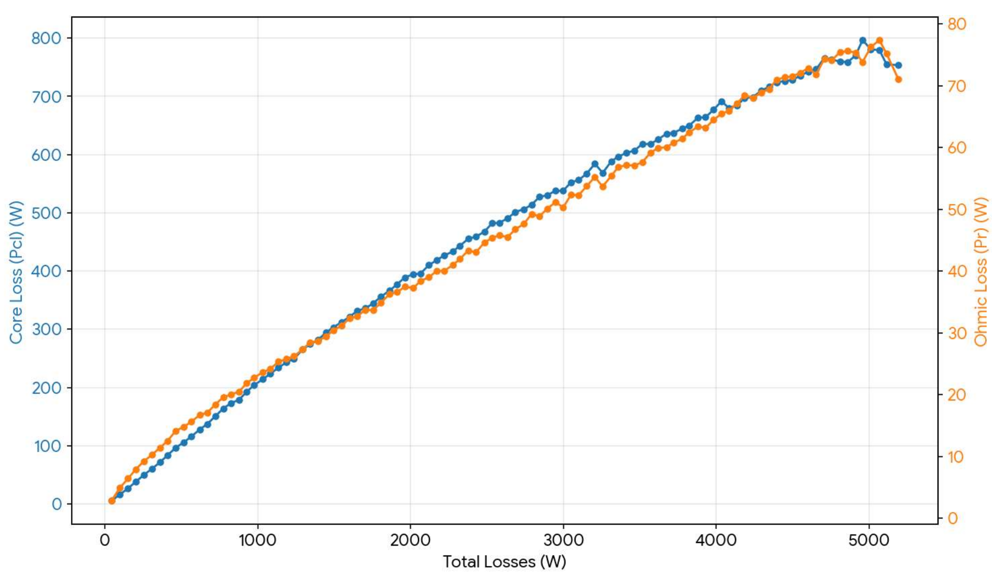

Fig 6. Variation of core loss and ohmic loss with respect to total full-load losses.

1.4 Validation and Interpretation

To confirm the validity and consistency of prediction for the resultant BLDC motor data, a hybrid descriptive statistics and supervised machine learning regression models' validation was conducted. In contrast to experimentally obtained datasets, this dataset is produced using deterministic analytical electromagnetic design equations within a structured Python-based computational environment. Thus validation consists primarily in demonstrating that the results are internally coherent, physically consistent, and predictively stable in the extended design space. In order to make sure the machine learning results represent true prediction capability but not overfitting or data leakage, a strict validation protocol was adopted. The dataset was split using 70:30 stratified train–test split and all models were trained only on the training part, test set was never used for training. Further, learning curves are obtained using a five-fold cross validation. The features were pre-processed and normalized to avoid the rule domination by large magnitude features. There are no overlapping, or replicate entries between the training and testing subsets, removing possibility of data leakage.

Table 9 presents a comparative performance evaluation of eight regression techniques viz. Linear Regression (LR), Support Vector Regression (SVR), Decision Tree (DT), Random Forest (RF), XGBoost, LightGBM, AdaBoost, and Extra Trees (ET) across four predicted outputs namely power input, rated output power, torque, and efficiency. A consistent observation across deterministic outputs (Power Input, Rated Output Power, and Torque) is the near-perfect performance of tree-based ensemble methods (Random Forest, Extra Trees, LightGBM, XGBoost), with R^2 values approaching 1.0 and extremely small MAE and RMSE values. Such behavior is to be expected since these parameters are calculated from explicit analytical expressions with no noise of measurement involved. The determinism in the dataset lets ensemble models capture nonlinear relations between geometric and electrical variables. The Linear Regression also attains perfect prediction (R^2=1.0) for line Rated Output Power and Torque for some parameters. This suggests that these variables are also stringently linearly dependent on the first design parameters in the devised analytic framework. On the other hand, the R^2 values on the efficiency prediction is a bit lower for most models (varying between 0.30 and 0.88). This non-conformity is conceptually significant; it is not a "bug" of the dataset. Efficiency is a ratio output that depends on the product of several nonlinear terms (core loss, copper loss, stray loss and output power), and small parametric perturbations are amplified. Thus, calculation of efficiency will always be more difficult than predicting variables immediately raised, such as torque or power.

Table 9
Performance comparison of regression models

Outcome	Model	MAE	RMSE	R^2
Rated Output Power (kW)	*LR*	*0*	*0*	*1*
	SVR	*0.9967*	*2.6214*	*0.9919*
	Decision Tree	*0.0017*	*0.0289*	*1*
	Random Forest	*0.0035*	*0.0164*	*1*
	XGBoost	*0.0388*	*0.0584*	*1*
	LightGBM	*0.0009*	*0.0013*	*1*
	AdaBoost	*3.1453*	*3.6222*	*0.9846*
	Extra Trees	*0.0046*	*0.0103*	*1*
Torque (N-m)	*LR*	*0*	*0*	*1*
	SVR	*6.5770*	*14.8473*	*0.9745*
	Decision Tree	*0.0088*	*0.1186*	*1*
	Random Forest	*0.0116*	*0.0552*	*1*
	XGBoost	*0.1253*	*0.1867*	*1*
	LightGBM	*0.0029*	*0.0041*	*1*
	AdaBoost	*10.0108*	*11.5299*	*0.9846*
	Extra Trees	*0.0149*	*0.0348*	*1*
Efficiency (%)	*LR*	*0.0526*	*0.0773*	*0.8516*
	SVR	*0.0466*	*0.0695*	*0.8800*
	Decision Tree	*0.1309*	*0.1674*	*0.3040*
	Random Forest	*0.0902*	*0.1175*	*0.6573*
	XGBoost	*0.0675*	*0.0923*	*0.7885*
	LightGBM	*0.0657*	*0.0907*	*0.7957*
	AdaBoost	*0.1206*	*0.1520*	*0.4259*
	Extra Trees	*0.0856*	*0.1141*	*0.6764*
Power Input (kW)	*LR*	*0*	*0*	*1*
	SVR	*1.0789*	*2.8251*	*0.9915*
	Decision Tree	*0.0336*	*0.0567*	*1*
	Random Forest	*0.0279*	*0.0491*	*1*
	XGBoost	*0.0781*	*0.1088*	*1*
	LightGBM	*0.0204*	*0.0281*	*1*
	AdaBoost	*3.2553*	*3.7347*	*0.9851*
	Extra Trees	*0.0221*	*0.0408*	*1*

1.4.1 Rated Output Power

The true vs predicted value plots display a nearly perfect fitting of predicted values with the 45-degree line for ensemble-based LightGBM, Random Forest, Extra Trees, Decision Tree. The very tight grouping of points suggest the bias is zero, and the scatter is minimal for the whole power range. The SVR shows a slightly bigger spread compared to ensemble trees, since it is very sensitive to the kernel parameter tuning. Nevertheless, the bias is still low and acceptable.

Residual plots of the ensembles show no apparent heteroscedastic pattern and are roughly symmetric around zero. There is no evidence of systematic curvature, indicating omission of nonlinear structure in the models. The magnitudes of the residuals are still very tiny compared to the rated power values, which again suggests accurate predictability.

The learning curves show high training error at the beginning, a steady decrease of the cross-validation error, and a small gap between the training-validation learning curves. Such behavior indicates no overfitting, good generalization, and sufficient dataset size. For deterministic results like rated output power, ensemble methods converge quickly because of evident parametric mapping. Fig. 7-9 illustrates the true vs predicted values, residual analysis and learning curves respectively.

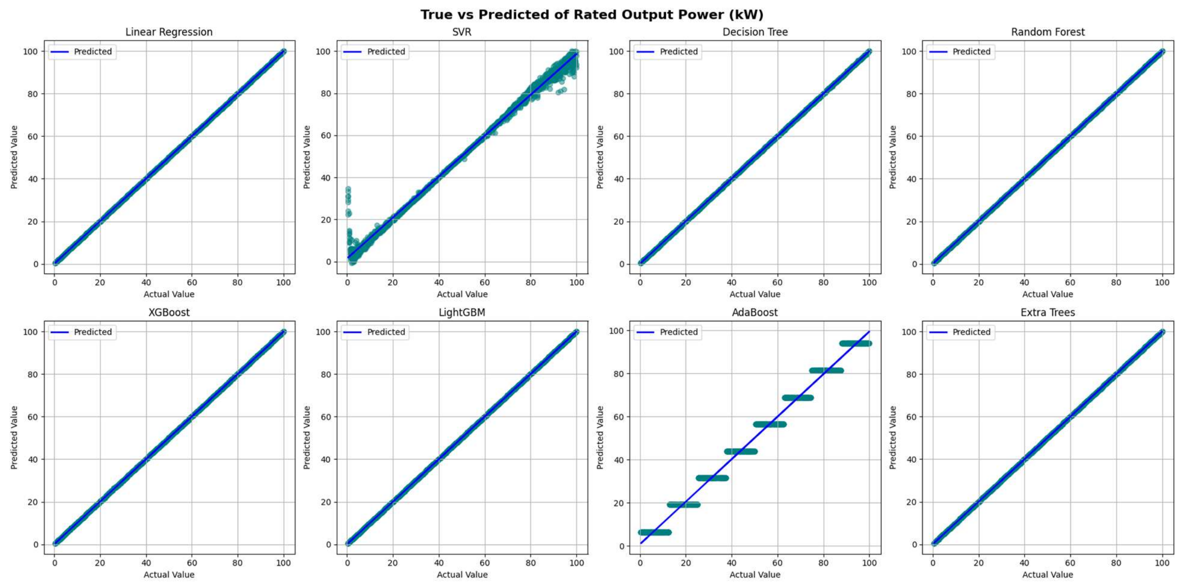

Fig 7. True vs predicted of various models for Rated Output Power (kW)

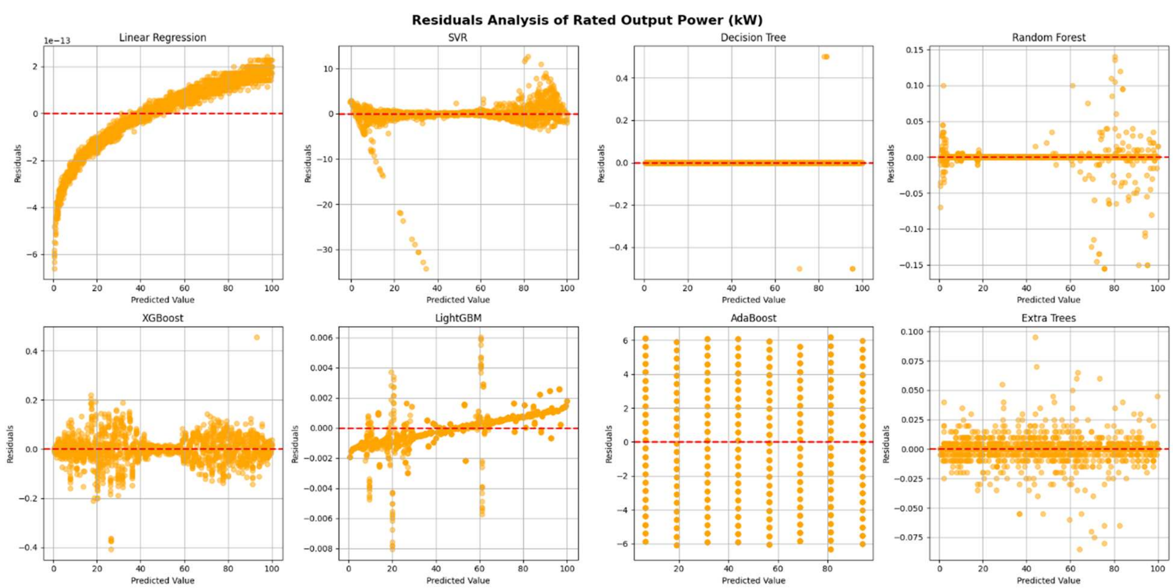

Fig 8. Residual analysis of various models for Rated Output Power (kW)

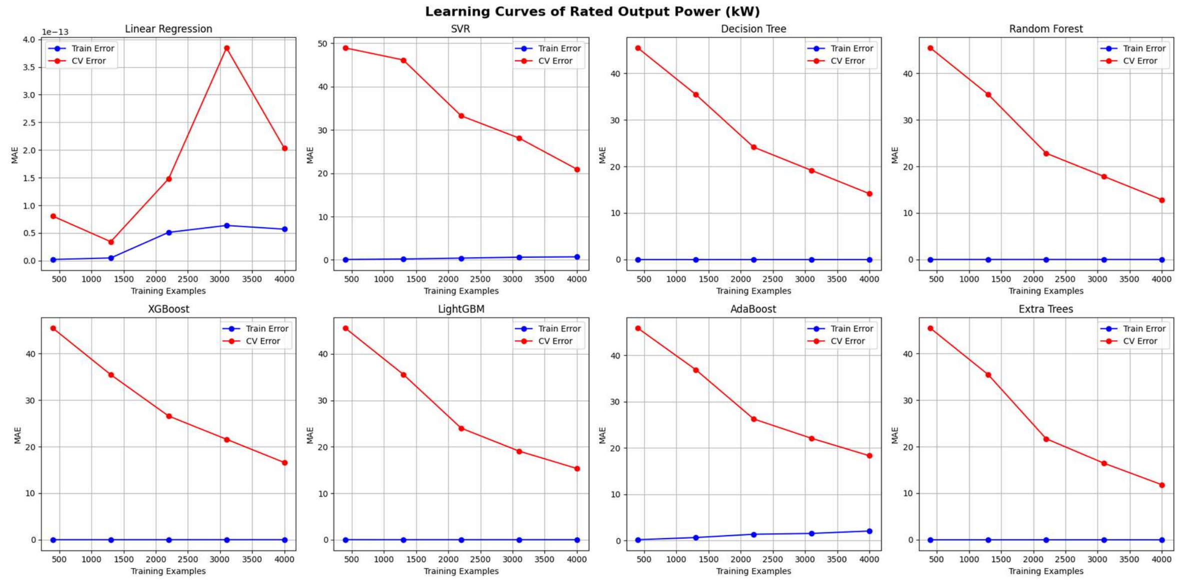

Fig 9. Learning curves of various models for Rated power output (kW)

1.4.2 Torque

Torque prediction demonstrates high fidelity across all torque bins, with negligible deviation from actual values. LightGBM, DT and RF are all with almost 100% adherence to the reference line. Since torque is functionally dependent on phase current and magnetic loading, its map is well organized and readily learnable. SVR exhibits a slightly higher variance for larger torque values, which is reasonable since kernel-based regression can potentially underfit regions with rapid changes in gradients when hyperparameters are not sufficiently tuned.

Ensemble models have residuals very tightly clustered to zero. No systematic bias is apparent for low or high torques. The lack of a funnel shape dispersion implies homoscedasticity and equal prediction accuracy in all bins of torque.

Training and validation curves quickly approach each other with little separation. This indicates a high signal-to-noise ratio in the data and parametric relations are stable and torque bins are sufficiently represented. The deterministic torque–current relationship is well represented by tree ensembles. Fig. 10-12 illustrates the true vs predicted values, residual analysis and learning curves respectively.

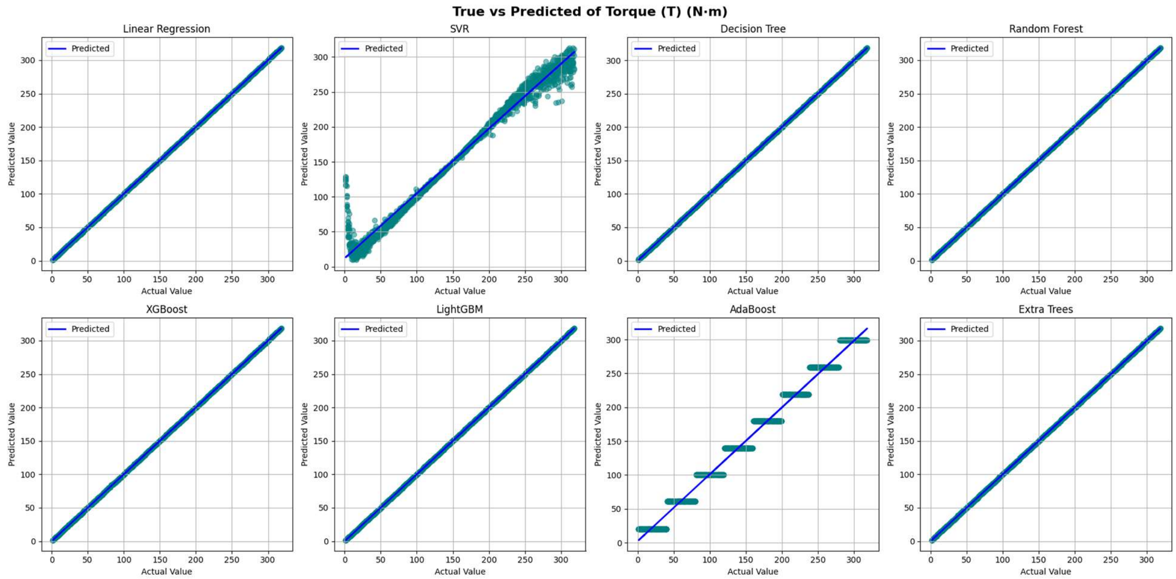

Fig 10. True vs Pred of various models for Torque (Nm)

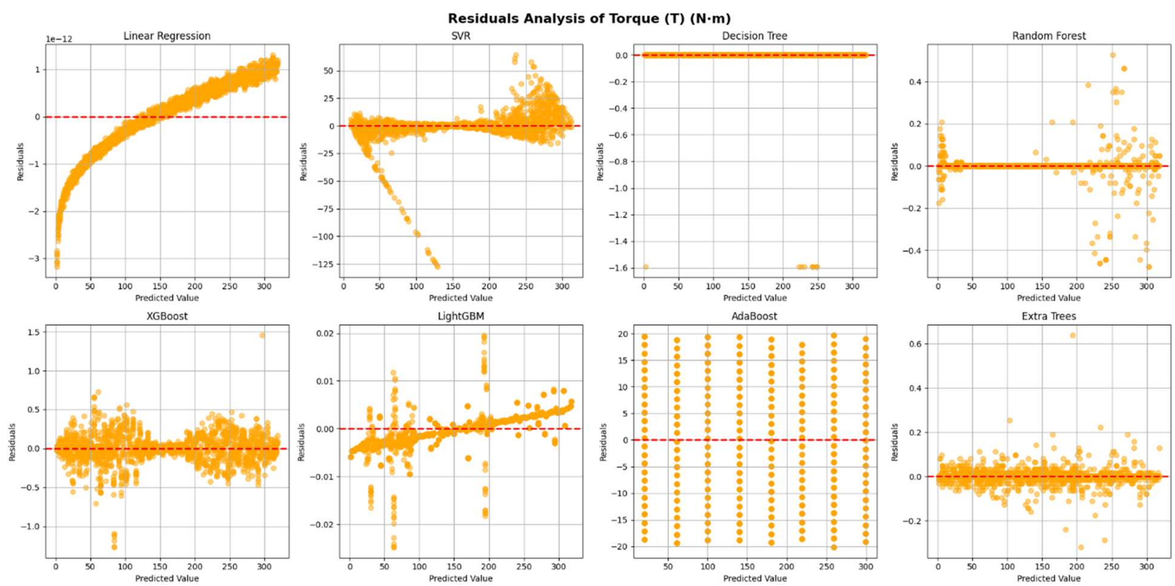

Fig 11. Residual analysis of various models for Torque (Nm)

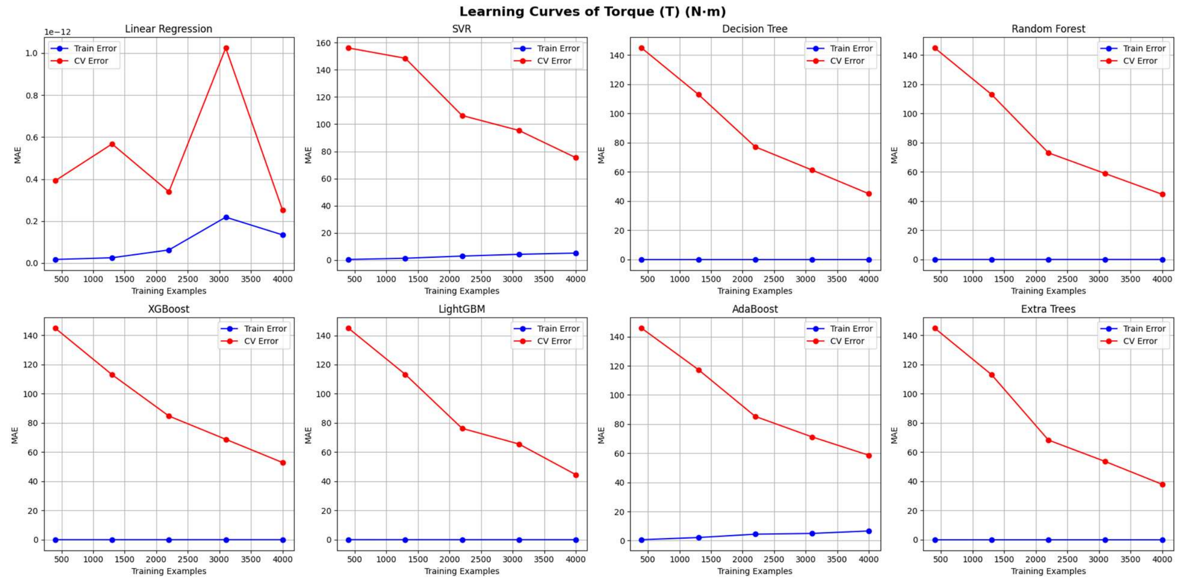

Fig 12. Learning curves of various models for Torque (Nm)

1.4.3 Efficiency

Unlike torque or power, efficiency is not directly computed from a single dominant variable but emerges from coupled loss mechanisms, resulting in comparatively higher prediction sensitivity. In contrast to the torque and power, the predictions for the efficiency can be seen to scatter around the reference at a certain distance. This is not a deficiency but a consequence of efficiency being computed from ratio relationships, influenced by both the numerator (output power) and the denominator (input power), and magnifying minor variations in losses. Among all the models SVR, LightGBM and Linear Regression consequently have less scattered results. Decision Tree exhibits significant scatter due to its piecewise constant representation, which is ill equipped to approximate smooth ratio-based functions.

Although the residuals for efficiency remain centered around zero, their dispersion is slightly higher compared to other predicted parameters, reflecting the greater nonlinear sensitivity of efficiency as a ratio-based metric. Importantly, no systematic bias trend is detected, no monotone drift is identified across bins and the dispersion remains statistically bounded. This rules out that the complexity of efficiency is caused by nonuniformity in the dataset.

Efficiency learning curves have slightly larger cross-validation error, show slow decreasing trends with larger training size and there are no signs of divergence in training and validation curves. This is interpreted as lack of drastic overfitting, efficiency need bigger sample diversity, complexity of model has influence on quality of prediction. The relatively lower R^2 for efficiency can therefore be attributed to an inherent mathematical sensitivity rather than a model-based weakness. Fig. 13-15 illustrates the true vs predicted values, residual analysis and learning curves respectively.

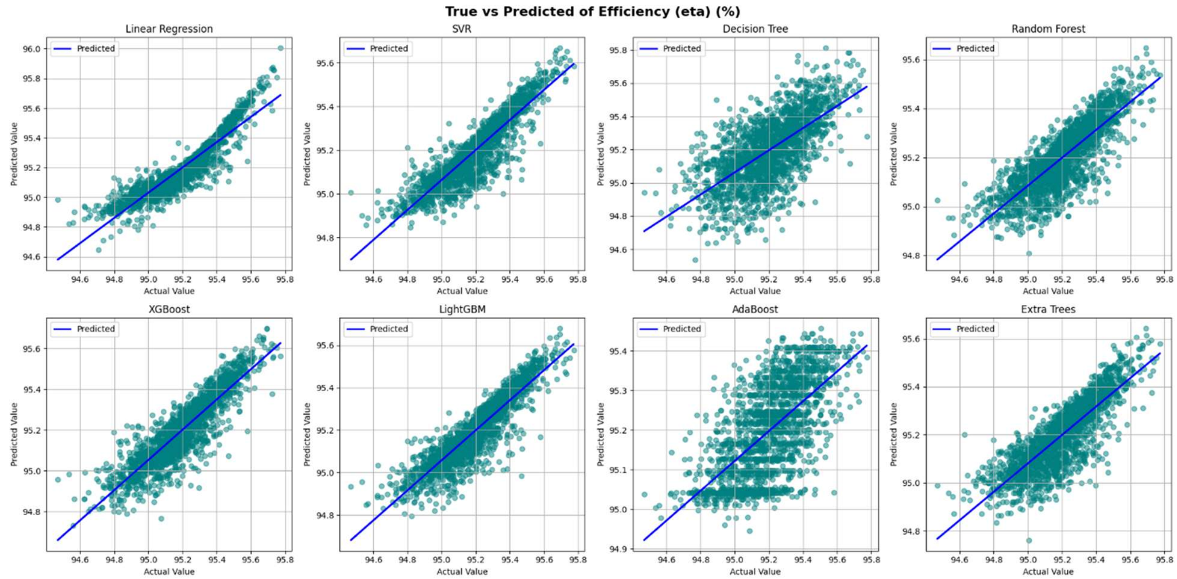

Fig 13. True vs Pred of various models for Efficiency (%)

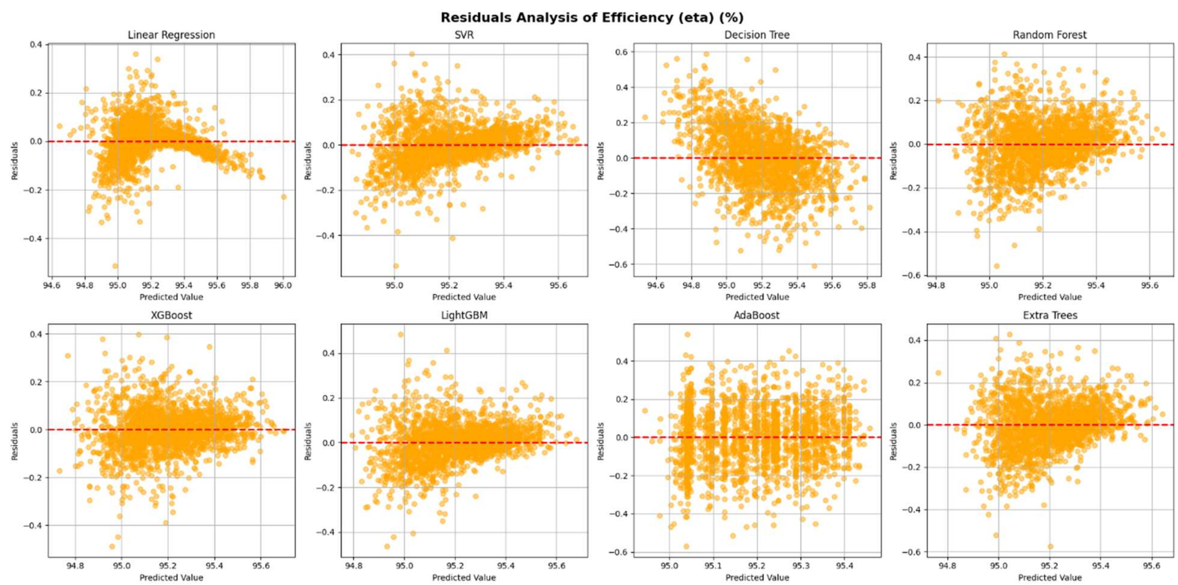

Fig 14. Residual analysis of various models for Efficiency (%)

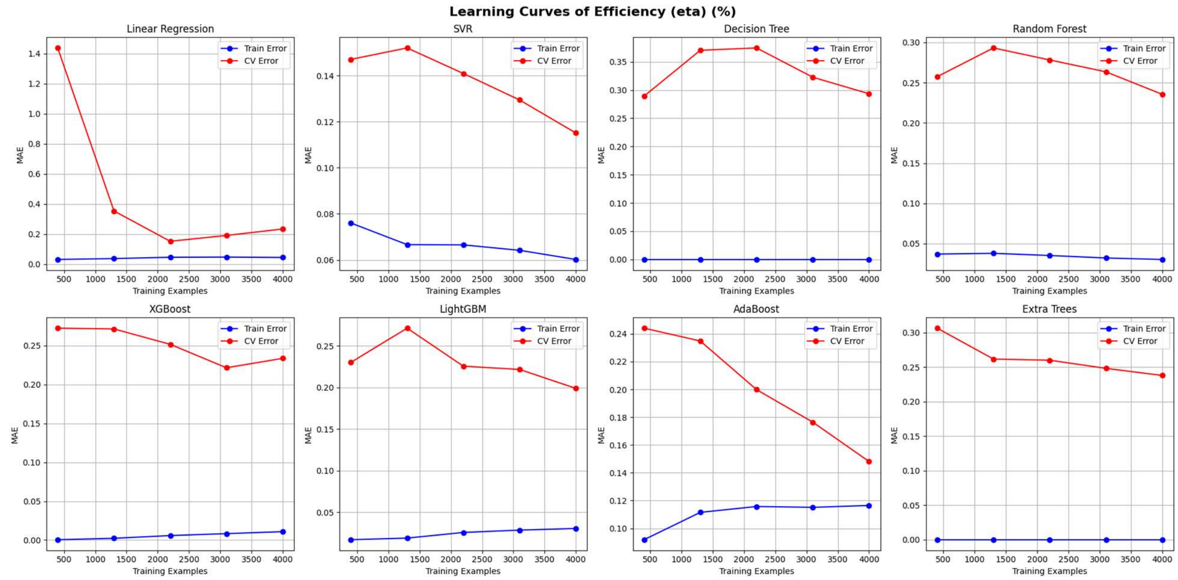

Fig 15. Learning curves of various models for Efficiency (%)

1.4.4 Power Input

The True-vs-Predicted plot for Power Input shows the predicted values are very closely aligned to the reference 45 degree line for the ensemble methods (LightGBM, Extra Trees, Random Forest and Decision Tree). This shows that the systematic bias is negligible and the model fidelity is excellent as the Very narrow band is maintained over full power range. Linear Regression also predicts exactly (R^2 = 1.0) which is physically feasible since the input power is based from analytical expressions of the torque and efficiency relations. This shows that Power Input still has a close to 1 mapping linearly within the dataset. SVR and AdaBoost have a relatively larger spread than ensemble trees but the differences are still relatively minor when compared to the magnitudes of the input power values. Taken as a whole the figure demonstrates that Power Input is an extremely learnable and structurally deterministic feature within the data set, as we suspected.

The distribution of residuals for Power Input is closely clustered around zero for all the best performing models. There are no apparent heteroscedascticity or curvature. The sizes of the residuals are still miniscule relative to the size of the predicted power values. Most importantly, no increasing spread with higher power bins, no systematic over-prediction or under-prediction, no structured residual patter was found. This confirms that the analytical relationships defining Power Input are consistently captured by ensemble models without bias.

The learning curves exhibit very low training error from early training stages, rapid convergence of cross-validation error and minimal separation between training and validation curves. This indicates no evidence of overfitting, strong generalization capability and sufficient dataset size (6000 samples adequate). The stability of learning curves further confirms that the deterministic nature of Power Input enables reliable regression performance. Fig. 16-18 illustrates the true vs predicted values, residual analysis and learning curves respectively.

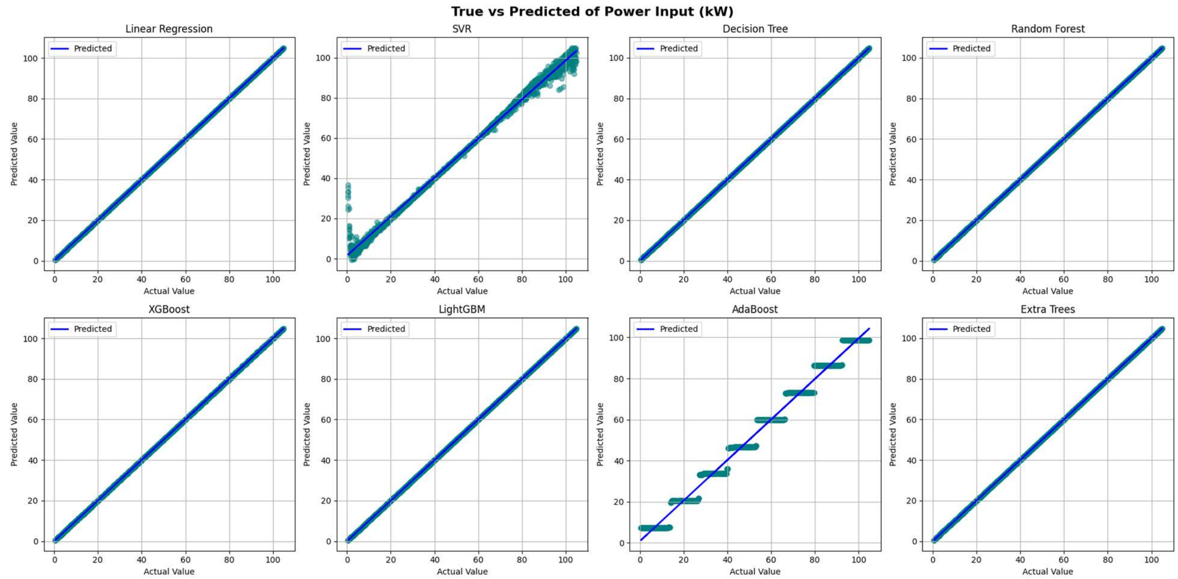

Fig 16. True vs Pred of various models for Power Input (kW)

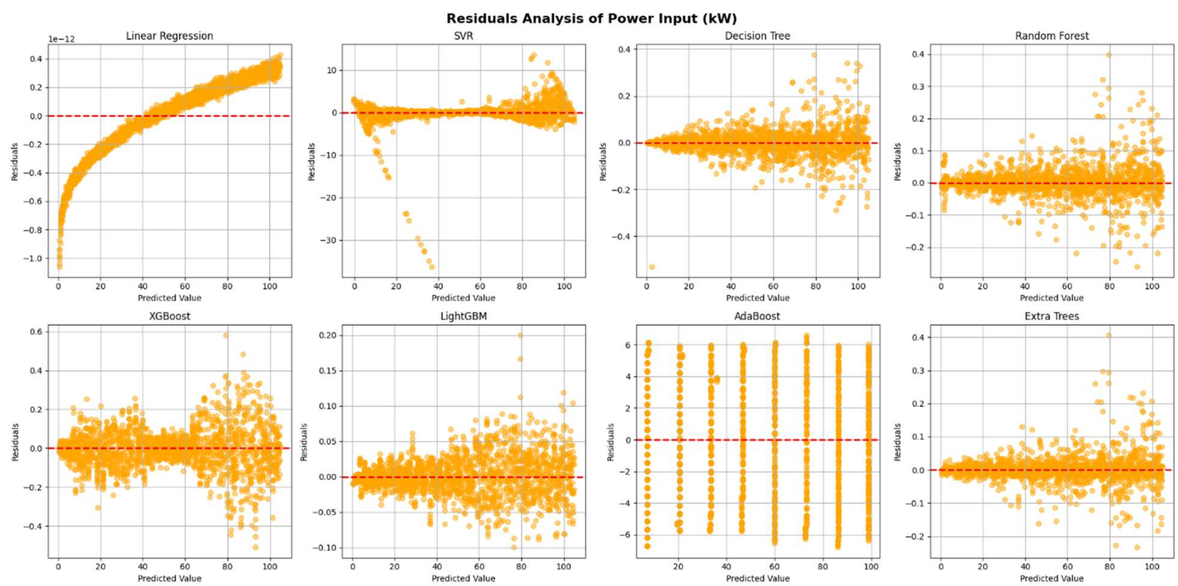

Fig 17. Residual analysis of various models for Power Input (kW)

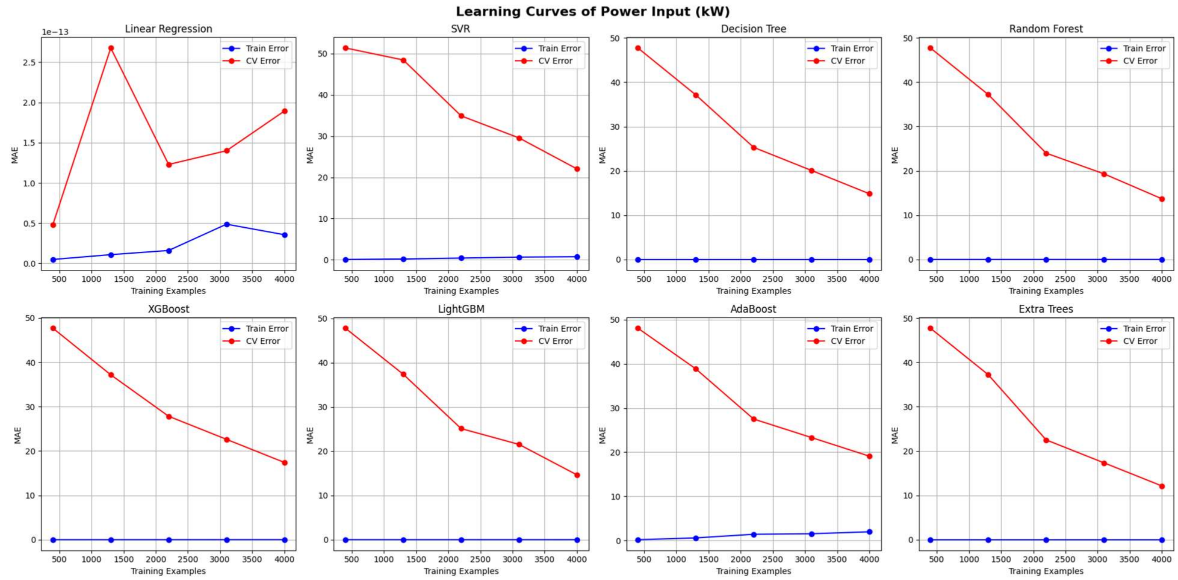

Fig 18. Learning curves of various models for Power Input (kW)

The comparative analysis of eight regression algorithms across four output parameters reveals that deterministic analytical outputs such as Rated Output Power, Torque, and Power Input are highly predictable with near-unity R^2 values across ensemble-based tree methods. This reflects the strong parametric structure embedded within the physics-based dataset. Residual analysis of each model suggests symmetric errors without bias, and learning curves indicate fast convergence with minimal overfitting. R^2 values for the prediction of efficiency are relatively lower, since the efficiency depends nonlinearly on several competing loss terms. SVR is the most accurate for predicting the efficiency among the considered models, while LightGBM is the most stable in general for all the parameters. Gradient boosting algorithms (LightGBM, Random Forest, Extra Trees) also shine in terms of robustness and stability, thanks to their ability to model non-linear interdependencies between geometric, electrical and performance metrics. The combined evidence suggests the created BLDC dataset is tightly self-consistent, structured, and appropriate for surrogate modeling and machine learning driven design.